SERVICE BASED ARCHITECTURES

ACIT3855 – WINTER 2026

COURSE OVERVIEW AND LEARNING OUTCOMES

This course focuses on the design of Enterprise Application Software using Service Based Architecture. Students use Service Based Architecture analysis and design techniques to identify business and systems services. The course will consider various techniques for implementing services including Web Services and Representational State Transfer. The course will review current and emerging standards that are relevant to Service Oriented Architecture, SaaS, and Micro-services.

Upon successful completion of this course, the student will be able to:

- Explain what Service Based Architectures are and their impact on modern Enterprise Application Software.
- Apply the concepts and principles of Service Based Architectures to software development projects.
- Compare and contrast Service Based Architecture analysis and design with object-oriented (OO) analysis and design.
- Design an interoperable Service Based Architecture application.
- Implement services using REST interfaces.

EVALUATION CRITERIA

Quizzes	10%	There will be an in-class quiz at the beginning of each class (except the first and last). Based on pre-reading materials posted to the Learning Hub.
Project Labs	35%	A microservices based application will be incrementally built and deployed through the project labs. Labs are done individually, typically require a demo and are due by the end of the next class.
Midterm Exam	20%	Concept and Coding, On Paper
Assignment	5%	One coding assignment at the end of the term
Final Exam	30%	Concepts and Coding, On Paper and Laptop

Quizzes will **only be done in-class only.** There are no late or make-up quizzes.

There is a 20% penalty per week for late lab submissions, up to a maximum of 2 weeks.

Passing Criteria:

- Students must complete a minimum of 80% of the Project Labs.
- Students must achieve a minimum of 50% on the combination of the Midterm and Final Exam (i.e., 25/50).
- Students must achieve an overall grade of at least 50% in the course.

COURSE SCHEDULE

Week	Topics	Notes
1	• Services Based Architecture Overview • RESTful APIs Review	Lab 1
2	• Microservices Overview • Edge Service	Lab 2, Quiz 1
3	• Database Per Service • Storage Service (SQLite)	Lab 3, Quiz 2
4	• Logging, Debugging and Configuration • Storage Service (MySQL)	Lab 4, Quiz 3
5	• RESTful API Specification (OpenAPI) • Processing Service	Lab 5, Quiz 4
6	• Synchronous vs Asynchronous Communication • Message Broker Setup and Messaging	Lab 6, Quiz 5
7	• Event Sourcing • Analysis Service	Lab 7, Quiz 6
8	• Midterm Week	Midterm Exam
9	• Deployment - Containerization of Services	Lab 8, Quiz 7
10	• Dashboard UI and CORS	Lab 9, Quiz 8
11	• Issues and Technical Debt	Lab 10, Quiz 9
12	• Deployment – Centralized Configuration and Logging	Lab 11, Quiz 10
13	• Deployment – Load Balancing and Scaling • Final Exam Preview	Lab 12, Quiz 11
14	• Review Week	Lab 12 Demos, Assignment Due
15	• Final Exam	

You will need to use a Cloud VM (AWS or Azure or Equvalent) starting Week 6 as we will start using Docker for deploying 3rd party services and your own services.

INSTRUCTORS

- Set A – Mike Mulder
- Set B – Rafi Mohammad
- Set C – Mike Mulder

Please contact the instructor for your set for questions and to demo your labs.

Our office hours is posted as a news item on the Learning Hub.

CLASS STRUCTURE

Pre-reading Before Class – On the week's tool/concept

In Classroom (30-60 minutes):

- Quick Review – On the reading
- Quiz – On the Learning Hub
- Discussion/demo on the week's lab and/or concept

Lab (remainder of class time)

- Demos of previous week's completed lab. This may also be done prior to class during office hours.
- Work on new lab.

DISCORD AND OFFICE HOURS

The invite for the Discord Server is posted in the course site on the Learning Hub. We will be using this for the following:

- Lab questions/discussions and screenshares outside of classtime
- Notifications (in addition to news items on the Learning Hub)
- Office Hours

CHECK-IN QUIZ

- Complete Quiz 0 on the Learning Hub now. It will take less than 5 minutes.
- It will not count for marks, but I will use it to check attendance.

REMEMBER THIS FROM ACIT 2515?

EntityManager
- _entities : Entity[]
+ add_entity(Entity entity) : int + get_entity(int id) : Entity + update_entity(Entity entity) + delete_entity(int id)

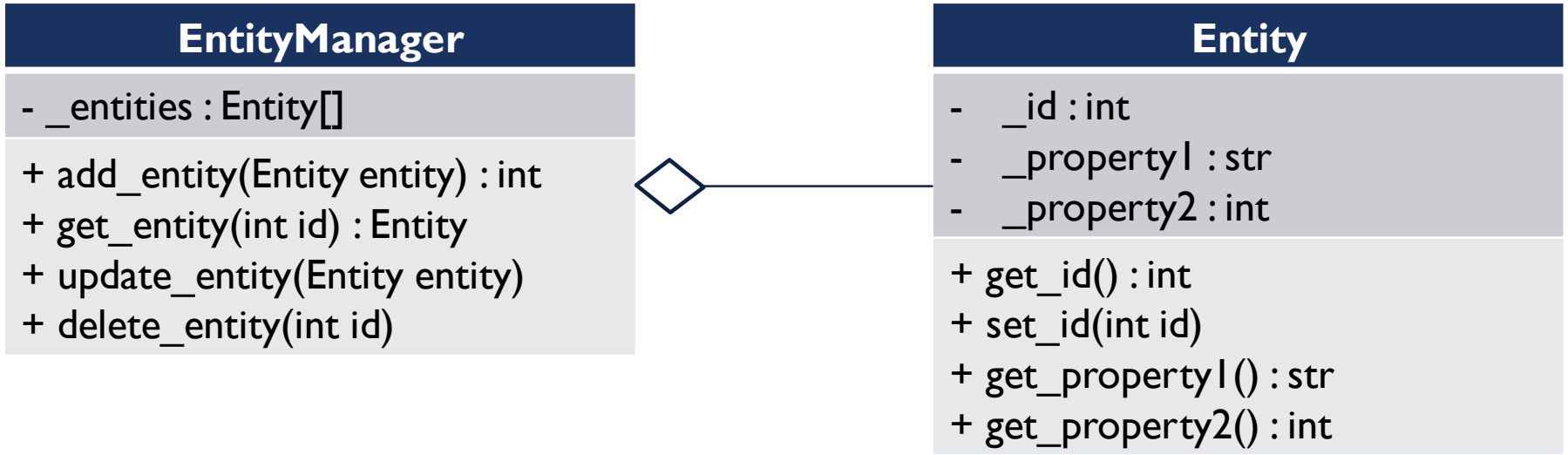
What do the methods in this class represent?

The Public Interface.
It can also referred to as an **API** – the Application Program Interface

Do you remember the Single Responsibility Principle (the S in SOLID)?

A class should only be responsible for a single part of the functionality of a software application. It can also be stated that the methods (the API) have <u>high-cohesion</u> – they are all related to the same task (i.e., they belong together)

REMEMBER THIS FROM ACIT 2515?



Different classes should be loosely coupled. They should only interact with each other through well-defined Public Interfaces (i.e., the API).

REMEMBER THIS FROM ACIT 2515?

EntityManager
- _entities : Entity[]
+ add_entity(Entity entity) : int + get_entity(int id) : Entity + update_entity(Entity entity) + delete_entity(int id)

POST /entities

GET /entities/<id>

PUT /entities/<id>

DELETE /entities/<id>

What was this?

The RESTful API we created on top of our EntityManager class

It provides an API to other software applications over a network using HTTP as the protocol.

SERVICE

Service

A service is a discrete unit of functionality that can be accessed remotely and can be acted upon and updated independently. *This is very much like our definition of a class but with RESTful API!*

A service typically has four properties:

- It logically represents a business activity with a specific outcome (i.e., single responsibility)
- It is self-contained
- It is a "black-box" for its consumers
- It may use other underlying services

SERVICE ORIENTED ARCHITECTURE

Service Oriented Architecture (SOA)

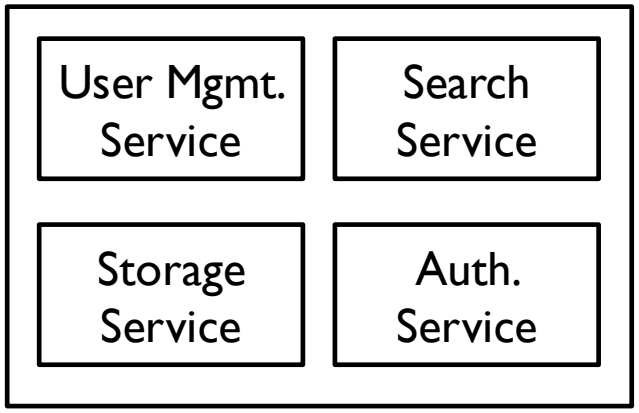
Style of software design where the software components are Services that interact through a communication protocol over a network.

Traditionally systems that use a SOA are decomposed into multiple software applications which are then further decomposed into services. Often these software applications are called **monoliths** because they consist of a single software deployment that is very large and complex.

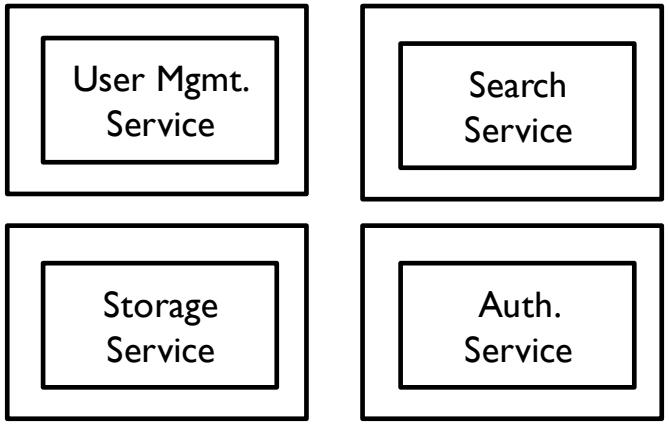
MICROSERVICES

Microservices

A form of SOA where the services are more fine-grained and independently deployable.



Monolith – Single Deployment



Microservices – Multiple Small Deployments

How Small is "Micro"?

MICROSERVICES

Advantages

- Breaks down the system into smaller pieces that can be worked on independently
- Enables ownership of individual services by specific developers or teams
- Allows for better scalability of the system – each service is deployed independently and can scale independently
- Allow for independent technology choices – each service could use a different programming language and related technologies

Disadvantages

- Allows for more diversity which can be harder to manage and make knowledge transfer more difficult
- It is much more difficult to deploy and operate multiple applications for a single system

RESTFUL APIS

Although microservices allow for diversity in technologies internally, generally they provide RESTful APIs to their clients externally.

What do you remember about RESTful APIs from ACIT 2515 (and other courses)?

RESTFUL APIS

- URI: Defines the endpoint
  - Example: POST http://api.example.com/users
- HTTP Resource (Noun): The data object you are operating on (i.e., user in the example above)
- HTTP Method (Verb): The operation you are performing on the resource (i.e., POST in the example above)
  - POST - create
  - PUT - update
  - GET – retrieve (no side effects)
  - DELETE – remove
- Request and Response Messages – Typically defined in a JSON format

RESTFUL API SPECIFICATION

- RESTful APIs are typically specified by the code. However, that means someone needs to read the code to understand how to use the API.
- Therefore, we want to document our API. However, if the documentation is separate from the code, it can get out of synch quickly.
- So strategies for specifying RESTful APIs that could be used to generate both documentation and code were developed. Two competing specifications were popular:
  - Swagger
  - OpenAPI
- Swagger and OpenAPI merged together a couple of years ago – now OpenAPI defines the specification for describing RESTful APIs and Swagger provides the tooling to generate documentation/code/etc. from the specification.

RESTFUL APIS VS OPENAPI

```
@app.route("/weather/temperature", methods=["POST"])
def report_temperature():
    data = request.json
    return "", 201
```

```yaml
paths:
  /weather/temperature:
    post:
      requestBody:
        required: true
        content:
          application/json:
            schema:
              required:
                - station_id
                - temperature_celsius
      responses:
        '201':
          description: Reading stored
```

OPENAPI 3.0 AND SWAGGERHUB

- SwaggerHub is an online service that allows you to specify your RESTful API using OpenAPI.
  - Provides validation of your RESTful API specification and automatically generates corresponding documentation
  - https://swagger.io/tools/swaggerhub/
- OpenAPI allows you to specify the resources and supported HTTP methods for each resource in your API
  - https://app.swaggerhub.com/help/tutorials/openapi-3-tutorial
- We will be using SwaggerHub and OpenAPI 3.0.x in this class to specify the RESTful APIs for our services.
- You will try these out in today's lab (Week 1) and we will go slightly deeper in Week 5.

COURSE PROJECT

Through the labs you are going to build a sample application using a Microservices Architecture.

You can define your application but it must have the following:

- The overall purpose of the system is to receive, process and store data for "Big Data" analytics.
- It receives batches of data from other systems periodically. These batches may be received throughout the day but have times when the number of concurrent data batches received can be very large such that the system may need to scale to handle the load.
- The data from the batch events are stored for later data analytics purposes.
- Some basic analytics and processing will be done in the system itself.

COURSE PROJECT – EXAMPLE 1 (MINE, DON'T USE)

This is a weather data collection system that receives temperature and precipitation readings from weather stations around the world. The weather stations batch up each type of reading and transmit them to the collection system periodically every 2 to 8 hours. Therefore, the number of batched readings received can get very high at the top of each hour, with as many as 1000 batches of each type (temperature or precipitation) received concurrently.

The readings are stored individually in the system to allow the following:

- Tracking short term weather trends for weather forecasting
- Tracking long term weather patterns and trends (i.e., for monitoring global warming)

Users of the system are meteorologists and climate scientists.

COURSE PROJECT – EXAMPLE 2 (SAMPLE ONLY, DON'T USE)

This is a system that receives and analyzes readings from medical devices at a patient's home (i.e., blood pressure, pulse, etc.) to allow for analysis of patient data across a large population of individuals with chronic health conditions.

The readings are the blood pressure and pulse readings (i.e., heart rate). The device stores the readings taken during the day and then sends a batch of each type to the system at midnight of each day (in the patient's timezone). Therefore, the number of batches received can be very high at the top of each hour, with a maximum of 10,000 received concurrently.

The readings are stored individually in the system to allow the following:

- Aggregation of data across multiple patients to look for trends in the data
- Allowing for predictive analytics on anomalies in patient data by correlating the readings with patient health records

Users of the system are data scientists and large health authorities.

COURSE PROJECT – EXAMPLE 3 (SAMPLE ONLY, DON'T USE)

This is a system for an ecommerce retailer who want to perform data analytics on its user data across its variable websites. Each website will track user location and click data and batch up the data. The batches of each type of user data will be reported to the system every 30 minutes to 1 hour depending on the amount of traffic on the site.

The number of batches will be higher at the top and bottom of each hour, but will be limited to less than 50. But the number of data items in each batch will be very high with as many as 10,000 records in each batch.

The requests are stored to allow the ecommerce retailer to perform data analytics such as:

- Tracking and prediction of user conversion to customer rates at each website
- Analyzing long term customer trends in terms of products clicked on and customer location

User of this data includes data scientists and marketing teams.

OPENAPI / SWAGGERHUB DEMO

- Let's look at an OpenAPI specification for the events for my system for Example 1 that includes endpoints for reporting:
  - Temperature Readings
  - Precipitation Readings

LAB TOOLS – DEVELOPMENT OF SERVICES

RESTful API Specification: SwaggerHub and OpenAPI

- Define a RESTful API in a yaml format

RESTful API Implementation: Python connexion

- Built on top of Flask but allows integration with an OpenAPI specification

RESTful API Testing: PostMan and Apache jMeter

- Postman – same as ACIT 2515
- Apache jMeter – for load testing

Today you will be defining the RESTful API your first service in OpenAPI.

Next class you will be building and testing this service.

In subsequent classes you will be building more services following a similar pattern.

TODAY'S LAB (LAB 1)

The lab is to be submitted individually. Today you will:

- Define your project
- Design the RESTful API for the service that receives the events
  - Using OpenAPI and SwaggerHub
  - Must have two POST endpoints and each should have an array of objects (i.e., the batch) plus some common properties
    - In your batch data items, it should have at least one numeric property

Lab 1 is due midnight prior to next class (submission, no demo). It is an input for Lab 2, so try to get it done early so you can get feedback.

We will now transition to the lab. I will be available to answer questions and/or do an initial review of your project.